Source-Level Reliability Estimation under Publisher-Disjoint Evaluation

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Abstract. Whether a news outlet can be trusted is conventionally judged by expert organisations, which weigh editorial practice, factual accuracy and transparency at the level of the outlet itself. Computational formulations of the same task, however, usually operate one article at a time: a classifier is fitted to pre-labelled items and then scored on held-out ones. We instead cast the task as prediction about the source. Our framework has two stages: transformer-based language models rate individual articles, and those ratings are then pooled to characterise outlets that were never seen during training. To keep the setting close to deployment, every split is publisher-disjoint, so no outlet contributes to both the training and the test side. Working with 19,476 English-language political news items drawn from 439 outlets and carrying NewsGuard ratings, exact accuracy rises from roughly 0.60 on single articles to 0.69 on outlets, and from 0.84 to 0.88 once a prediction one level away from the true rating is counted as correct, a tolerance the labels invite because the five levels are bins of a continuous 0–100 score. We then ask how the errors distribute across the political spectrum, and find systematic associations between the leaning of an outlet and the mistakes made on its articles. Taken together, the results indicate that text alone carries enough signal to place an unseen outlet on a reliability scale, which supports content-based approaches to assessing sources at scale.

Keywords: News source reliability · Publisher reliability inference · Transformer-based text classification · Cross-publisher generalization · Reliability aggregation · Political bias analysis.

1 Introduction

Assessing the reliability of news publishers has become increasingly important for limiting the spread of misleading or low-quality information. Professional organizations such as NewsGuard, the Global Disinformation Index, and Media Bias Fact Check evaluate news outlets according to journalistic standards including transparency, editorial accountability, factual accuracy, and correction practices, and despite methodological differences they exhibit substantial convergence [11,19]. Importantly, these assessments are performed at the level of the publisher rather than of individual articles.

A large body of computational work instead formulates the problem as article-level classification [1,14,21,24]. While effective in controlled settings, this only partially reflects the structure of professional assessment, where reliability is a persistent property of the source emerging across multiple publications. Individual articles provide partial and noisy evidence: highly reliable outlets occasionally publish lower-quality content, and unreliable publishers produce articles that appear plausible in isolation.

A second limitation concerns evaluation. Existing studies derive article labels from publisher-level annotations without enforcing publisher-disjoint splits [1,6, 23], so models can partially memorize publisher-specific patterns rather than learning transferable indicators of reliability [3,26].

We address these limitations by reformulating the task as publisher-level inference. We propose a two-stage framework in which transformer-based language models first estimate the reliability of individual articles and then aggregate those predictions to infer the reliability of previously unseen publishers, under a strict publisher-disjoint evaluation protocol. Our contributions are fourfold: we reformulate news reliability assessment as a source-level inference problem; we introduce a publisher-disjoint protocol that prevents source leakage and measures generalization to unseen publishers; we construct a dataset of 19,476 political news articles from 439 English-language publishers labeled with NewsGuard scores; and we show that aggregating article-level predictions improves reliability estimation, reaching 0.69 publisher-level accuracy on unseen sources. Section 6 states the societal need addressed, the preconditions on which any benefit depends, and the harms the approach may introduce.

2 Related Work

Early work on online news quality focused on article-level classification, detecting misleading content through linguistic features or neural architectures [14], often without considering the publisher.

Publisher reliability and the signals used to estimate it A parallel line shifts the focus to the source, variously termed *reliability*, *trustworthiness* or *credibility* [8, 15,27], whose reputation is known to influence perceived information quality [9]. Computational approaches rely on textual and stylistic features [21] or on external signals such as user interactions, social networks and metadata [4,13,20,28]; journalistic rating systems are commonly used as ground truth [11,19], and recent work explores LLM-based estimation [10,18,24,29].

Datasets and cross-domain generalization Datasets with explicit publisher reliability labels remain limited [1,5,23], and recent work shows that performance degrades under distribution shifts across domains and publishers [3,6,26].

Our contribution In contrast, we focus exclusively on textual content and enforce a strict publisher-disjoint evaluation protocol.

3 Problem Definition and Methodology

3.1 Problem definition

We formulate the problem as 'News Source Reliability Inference': estimating the reliability level of a publisher from its published content. Because publishers are observed through collections of articles, we decompose it into (i) article-level reliability estimation and (ii) publisher-level inference through aggregation.

Let A be a corpus of news articles, each with a text in $\mathcal{T}$ and a publisher $p(a) \in P$, and let $A_p = \{a \in A \mid p(a) = p\}$ be the articles of p. Each publisher has a fixed reliability level $r_P(p)$ over five ordered categories $L = \{l_1, \ldots, l_5\}$ obtained by binning continuous scores (Section 4); since reliability is a publisher's property, every article inherits it, $r_A(a) = r_P(p(a))$. A political leaning $o_P(p) \in O$, with O ranging over Far Left, Slightly Left, Non labelled, Slightly Right and Far Right, is inherited in the same way. Both inheritance functions are needed because article-level annotation is unavailable; political leaning is never a model input or target, and is retained only for the post-hoc analysis of Section 5.4.

In **Stage 1** we train a classifier $C_{\text{reliability}} : \mathcal{T} \to L$ to predict the reliability of a single article from its text alone. In **Stage 2** we aggregate those predictions through $Agg : L^{|A_p|} \to L$, so that $\hat{r}_P(p) = Agg(\{C_{\text{reliability}}(a) \mid a \in A_p\})$; Section 5.5 instantiates Agg as the mode, i.e. majority voting.

Because each article inherits its publisher's label, articles from the same source are not independently labeled, and a random split at the article level would leak test labels into training. With P_{train} and P_{test} the publishers of the training and test articles, we therefore enforce

$$P_{\text{train}} \cap P_{\text{test}} = \emptyset. \tag{1}$$

All test articles thus originate from previously unseen publishers, which aligns the evaluation with the source-level nature of the labels and prevents publisher-specific memorization.

3.2 Evaluation Metrics and Baselines

We use the same metrics for both article-level and source-level evaluations.

Standard and Macro Metrics We report Accuracy as the ratio of correct predictions. Because class frequencies differ moderately (l_4 is about 2.6 times larger than l_5), we prefer Macro-averaged Precision, Recall and F1, which treat all five classes equally and prevent a model from appearing successful on the most frequent categories at the expense of minority ones.

Tolerant Metrics Given the ordinal nature of our labels, standard accuracy may be overly punitive for minor deviations. We therefore adopt *Tolerant Accuracy* [17], which assigns partial credit to predictions falling into adjacent categories. Let N be the total number of samples and c_{ij} the count of instances

with true label i predicted as j:

$$\text{Accuracy}_{\text{tolerant}} = \frac{1}{N} \sum_{i \in L} \sum_{j \in L} w_{ij} \cdot c_{ij}, \qquad w_{ij} = \begin{cases} 1 & \text{if } |i-j| = 0 \\ \alpha & \text{if } |i-j| = 1 \\ 0 & \text{if } |i-j| > 1 \end{cases} \tag{2}$$

The parameter $\alpha \in [0, 1]$ controls the tolerance level. We report results for $\alpha = 0.5$ (half credit for adjacent errors) and $\alpha = 1.0$ (full credit, effectively treating adjacent classes as correct).

Baselines We pair each baseline with the metrics it naturally challenges. The **Majority Baseline** assigns every instance to the most frequent class observed in the training set, and achieves the maximum accuracy obtainable without processing the input text. The **Uniform Random Baseline** assigns labels uniformly at random, giving each of the five reliability levels an equal probability of 20%. We use it for Macro metrics, which treat all classes equally and for which a Majority classifier would be an unsuitable benchmark, as it would score zero Recall and F1 on all other categories. It establishes a floor of 0.20.

3.3 Bias Assessment and Statistical Analysis

Beyond aggregate performance, we investigate *how* and *why* models fail.

Error Analysis We compute the Error Rate per level, $(1 - \text{Recall}_l)$. To identify systematic over- or under-estimation we analyze the Mean Error Direction (MED) over misclassified samples:

$$\text{MED} = \frac{1}{|E|} \sum_{i \in E} (\hat{y}_i - y_i) \tag{3}$$

where $\hat{y}_i$ and y_i represent the predicted and true reliability levels (mapped to their ordinal values $1, \ldots, 5$), respectively. Positive values indicate a tendency to overestimate reliability, while negative values indicate a tendency to underestimate reliability.

Statistical Significance of Political Bias To determine whether classification errors depend on the publisher's political leaning, we conduct a Pearson's χ^2 test of independence. We test the null hypothesis (H_0) that error frequency is independent of political orientation.

To quantify the magnitude of this effect independent of sample size, we compute Cramér's V:

$$V = \sqrt{\frac{\chi^2}{N \cdot \min(r-1, c-1)}} \tag{4}$$

where N is the total number of samples, while r and c represent the number of rows (political orientations) and columns (classification outcomes) in the contingency table.

4 Dataset and Labeling

Ground Truth: NewsGuard The publisher-level reliability function $r_P : P \to L$ is derived from NewsGuard,[1] a professional media rating system in which trained journalists evaluate online news outlets. Each publisher $p \in P$ is assessed against nine widely adopted journalistic criteria, including the avoidance of deceptive headlines, the transparency of ownership and financing, and the adoption of effective correction practices.

The evaluation produces a continuous reliability score in the range $[0, 100]$, which provides a quantitative measure of a publishers adherence to professional journalistic standards.[2] The proprietary ratings were licensed for use to the authors of this publication.

Label Construction and Publishers Selection We discretize the continuous score into five ordinal categories $L = \{l_1, \ldots, l_5\}$ following NewsGuard's own thresholds. Table 1 reports the mapping, with the number of publishers per label after filtering.

Label	Score Range	Initial Dom.	Scraped Dom.	Final Art.
l_1	0–39 (Max Caution)	~~1,211~~ → 175	152	4,871
l_2	40–59 (Caution)	127	69	3,016
l_3	60–74 (Credible w/ Exc.)	124	95	4,620
l_4	75–99 (Generally Credible)	175	88	5,041
l_5	100 (High Credibility)	68	35	1,928

Table 1. Reliability labels and NewsGuard score ranges, with the number of domains after filtering, successfully scraped domains, and collected articles.

We restrict P to outlets tagged as *Political News and Commentary*, publishing mainly in English and without a paywall. To prevent the corpus from being dominated by highly unreliable sources, we cap l_1 at 175 publishers, matching l_4; the final set contains $|P| = 669$ outlets.

Article Collection Let A denote the set of collected news articles. For each publisher $p \in P$, we scrape articles from its homepage using the `newspaper3k` library[3], collecting at most 200 articles per outlet to ensure a balanced representation across publishers. The scraper is configured to retrieve English-language content only and to prevent duplicates. Scraping did not succeed for all outlets, owing to anti-scraping measures, timeouts or structural incompatibilities, so the final publisher count is lower than the original selection. To exclude non-news or

[1] https://www.newsguardtech.com/
[2] https://www.newsguardtech.com/ratings/rating-process-criteria/
[3] https://pypi.org/project/newspaper3k/

low-information content, we retain only articles with a minimum length of 100 words, filtering out short announcements and advertisements. This procedure yields an initial corpus of 25,822 articles.

Data Curation and Labeling To preserve the original linguistic characteristics of the articles, we apply no text normalization or cleaning. Despite the scraper being configured for English, some domains returned localized content. We therefore filter out non-English articles using the `langdetect` library[4]. This process removes 6,346 articles and leaves 19,476 English-language articles in the dataset.

The final composition is given in the last two columns of Table 1; l_5 remains the least represented class, reflecting its limited presence in the original NewsGuard data.

Political Leaning Annotation NewsGuard also provides a political leaning annotation for a subset of publishers, inherited at the article level as $o_A(a) = o_P(p(a))$. Reliability levels are unevenly represented across political categories, a structural property of the media ecosystem that must be considered when interpreting model performance and error patterns in the analyses that follow.

5 Experiments

This section presents the experimental setup and the results obtained for the tasks defined in Section 3.1: (i) *Article-level reliability estimation* (Stage 1) under a publisher-disjoint evaluation protocol, and (ii) *Publisher reliability inference* (Stage 2) obtained by aggregating article-level predictions. Model performance is assessed using the metrics introduced in Section 3.2.

5.1 Experimental Setup

Following the validation protocol described in Section 3 (Equation 1), we use a 5-fold cross-validation scheme with strict separation of publishers between training and test sets. This evaluation setting assesses the model's ability to generalize to articles from previously unseen publishers and approximates a realistic *in-the-wild* deployment scenario. Folds are stratified on publisher reliability, so that each preserves the same reliability distribution and the same proportions of publishers across reliability classes.

We select four transformer-based models spanning standard encoder baselines, optimized encoders and decoder-based large language models. **BERT**-base-uncased [7] is our primary baseline, establishing a lower bound comparable with a vast body of prior work [22]. **RoBERTa**-large [12] tests the limits of the encoder-only paradigm, verifying whether optimized training strategies alone improve reliability detection with the architecture unchanged. **Llama 3** 8B [25] represents the shift toward decoder-only generative models, assessing whether

[4] https://pypi.org/project/langdetect/

the parameter scale of a general-purpose LLM translates into superior classification accuracy. **Mistral** 24B [16], roughly three times larger than Llama, tests whether additional capacity improves detection or whether performance saturates.

Training Strategies and Hyperparameters Encoders (BERT, RoBERTa) were *fully fine-tuned* with AdamW, a learning rate of $2e^{-5}$ and weight decay [7,12]. For the decoders (Llama, Mistral) full fine-tuning was computationally infeasible, so we adopted Low-Rank Adaptation (LoRA), freezing the pre-trained weights and updating only low-rank matrices injected into the attention layers, with a learning rate of $2e^{-4}$ in mixed precision.

Each model was trained for 5 epochs per fold, with the exception of Mistral, which was trained for 3 epochs. This duration was established following a preliminary convergence study (see Supplementary Material [2], which also reports the remaining optimisation and batching details), which demonstrated that validation loss minimizes at Epoch 4 for Llama, while Mistral exhibits comparable rapid convergence within three epochs. We used early stopping on validation loss and a fixed global random seed (42). The maximum sequence length for input processing was set to 256 tokens, except for Mistral, for which it was increased to 1,024 tokens. This tests whether a longer context improves classification.

All fine-tuned models and code will be released to foster further research; they are withheld during review to preserve anonymity and will be made available upon acceptance.

5.2 Stage 1: Article-level reliability estimation

We first evaluate the article-level estimators that constitute the first stage of the framework.

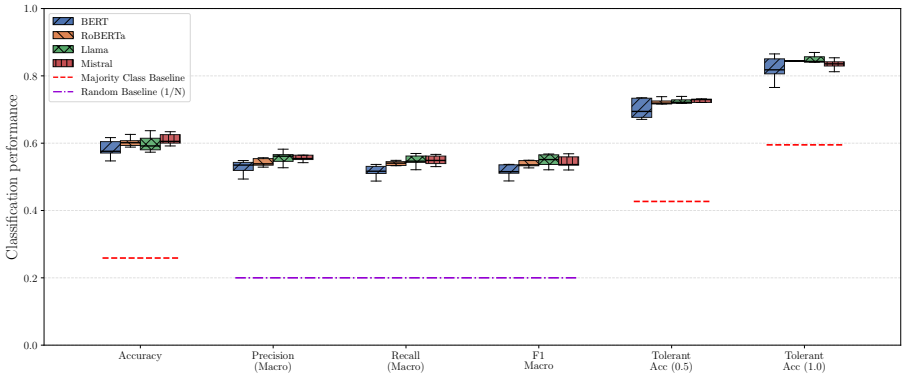


Fig. 1. Article-level metrics across the five folds. Horizontal lines are the baselines: dashed red for Majority Class, dash-dotted violet for Random.

Figure 1 illustrates the results against the Majority Class Baseline (MCB), used for Accuracy and Tolerant Accuracy, and the Random Baseline (RB) at $1/N$, used for the macro metrics. All models reach a standard accuracy around 0.60, more than double the MCB, with Mistral highest at 0.61. On macro-averaged performance RoBERTa leads with $F1_{macro}$ of 0.55, followed by Mistral and Llama at 0.54 and BERT at 0.52, all well above the RB of 0.20; the variance across folds is minimal (standard deviation about 0.02). Under the stricter tolerance ($\alpha = 0.5$), Mistral, RoBERTa and Llama reach approximately 0.72 against an adjusted MCB of 0.43. With $\alpha = 1.0$ RoBERTa reaches 0.84 and the others 0.82–0.83; the baseline is naturally higher here (0.60), so the gain is +0.24. Misclassifications are therefore rarely severe, typically involving adjacent labels.

5.3 Biases in Article-level reliability estimation

Here, we study whether the classifier systematically performs better on some reliability classes than on others.

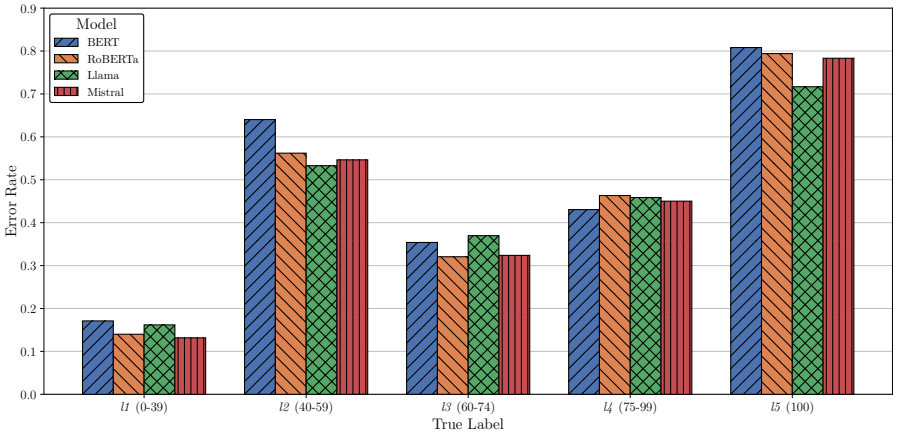


Fig. 2. Per-class error rate, $(1 - \text{Recall}_l)$.

Figure 2 highlights an asymmetry in prediction correctness. The lowest error rates are on l_1 (error rate < 0.20). This suggests that highly unreliable content has recognizable structural features. The highest-reliability articles (l_5) are the most challenging to recognize for the model, with error rates exceeding 70%[5]. Class l_2 also proves difficult, with an error rate of over 50%, while l_3 sits in the middle at approximately 35%.

[5] Class l_5 only includes articles with a reliability score of 100, and is easily confused with l_4, which contains similarly high scores.

Model	Accuracy	χ^2	p-value	Cramérs V
BERT	58.27%	1684.13	$< 10^{-300}$	0.294
RoBERTa	60.34%	1418.31	$< 10^{-300}$	0.270
Llama	59.97%	1207.83	$< 10^{-250}$	0.249
Mistral	61.16%	1343.98	$< 10^{-300}$	0.263

Table 2. Political leaning vs. classification correctness: χ^2 test of independence with p-values, and Cramér's V as effect size.

These patterns remain consistent across all four models, including the substantially larger Mistral, suggesting that the biases are structural properties of the task rather than intrinsic to a specific architecture or model scale; the Supplementary Material [2] reports the error-direction analysis and the full confusion matrices.

5.4 Influence of Political Leaning on Classification Errors

We investigate whether the models treat all sides of the political spectrum equally or penalize specific orientations. Political orientation is a property of the publisher, inherited by each article it produces (see Section 3.1), which enables a post-hoc analysis of how source-level alignment relates to article-level errors.

Because political orientations are unevenly represented in the dataset, we compute error rates separately for each leaning, which makes them independent of class frequency.

All models exhibit systematic variation across political leanings: there are substantially higher error rates for far left and far right outlets (5972%) compared to lower rates for slightly right and non-labelled outlets (3034%), and intermediate error rates for slightly left articles (4758%).

A chi-square test of independence rejects, for every model, the hypothesis that correctness is independent of political leaning (Table 2), with Cramér's V indicating a consistent small-to-moderate effect size.

We then focus on misclassified instances to assess whether models exhibit directional biases, namely systematic tendencies to overestimate or underestimate article reliability as a function of political leaning. Figure 3 reports the Mean Error Direction (MED), defined as the average signed difference between predicted and true reliability labels over misclassified instances for each orientation (see Section 3.3). As anticipated in Section 5.3, positive MED values indicate overestimation of reliability, whereas negative values indicate underestimation.

Clear directional patterns emerge across all four models. Far Left and Far Right articles exhibit consistent overestimation, with mean signed errors ranging from +0.33 to +1.00, approaching a full reliability level. Slightly Left articles show systematic underestimation, with average values between -0.54 and 0.82. Slightly Right and non-labelled outlets display smaller, less consistent deviations: Llama, RoBERTa, and Mistral slightly underestimate Slightly Right reliability

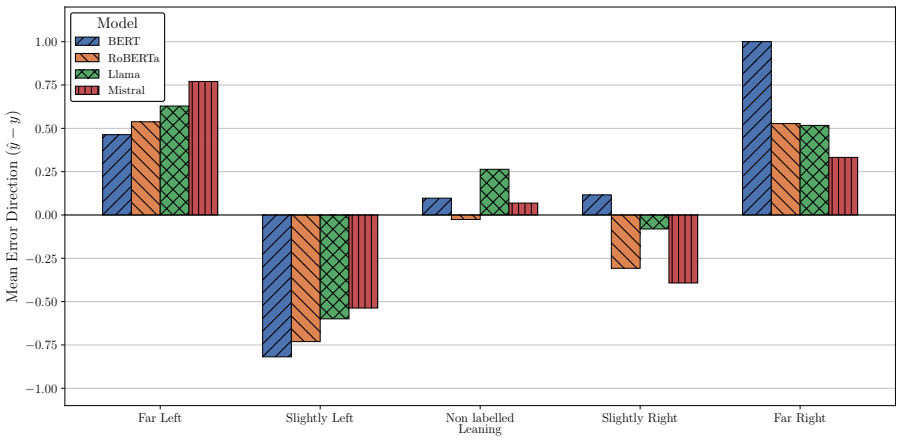


Fig. 3. Mean value and direction of misclassifications, stratified by political leaning. Positive values indicate overestimation (prediction > true label), while negative values indicate underestimation.

(up to -0.39), whereas BERT slightly overestimates it; for non-labelled outlets, BERT, Llama, and Mistral exhibit mild positive biases, while RoBERTa's average bias is near zero; the Supplementary Material [2] reports the corpus composition by leaning, the per-leaning error rates and the full confusion matrices stratified by political leaning.

5.5 Stage 2: Publisher reliability inference

Publisher-level inference is the primary objective of this work: the article-level estimators of the previous section are intermediate components rather than the final target. We instantiate *Agg* as the mode of a publisher's article-level predictions, i.e. majority voting, assigning the most frequent class as $\hat{r}_P(p)$, on the hypothesis that combining predictions offsets the noise of individual classifications. The analysis pools all article-level predictions across the five test folds, so that each publisher's reliability is inferred exclusively from articles the model never saw during training.

Figure 4 presents the results against the same two baselines used at the article level; the majority-class reference differs across the three accuracy variants, since a constant predictor also collects partial and then full credit under tolerance.

With an Accuracy of 0.69 RoBERTa leads, followed closely by Mistral at 0.68, more than triple the Random Baseline. Considering $\text{Accuracy}_{\text{tolerant}}$ (1.0), Llama achieves the top score of 0.88, followed by RoBERTa (0.87) and Mistral (0.86), substantially outperforming the majority class baseline.

Publisher-level errors are strictly local and concentrate between adjacent reliability levels, most notably between l_4 (Generally Credible) and l_5 (High

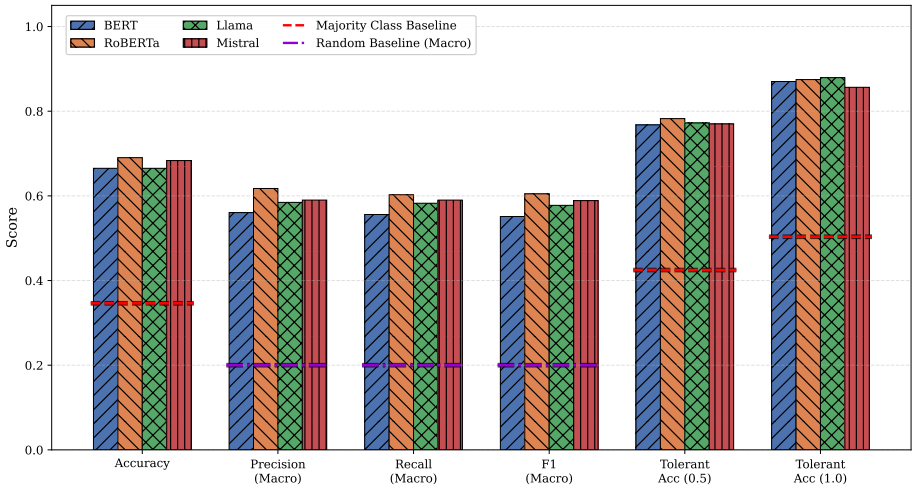


Fig. 4. Publisher-level metrics. Dashed red: Majority Class Baseline; dash-dotted violet: Random Baseline at 0.20.

Credibility); the Supplementary Material [2] reports full publisher-level confusion matrices and a detailed error analysis.

6 Theory of Change

Work on misinformation is not socially beneficial by default, and we therefore state what this work is expected to change, what must hold for it to do so, and how it may cause harm.

Societal Need and Intended Contribution Reliability assessments are produced by small teams of trained journalists, a procedure that is sound but cannot cover tens of thousands of outlets: the binding constraint is the cost of *triage* rather than the quality of expert judgement. We show that article text alone places an unseen outlet on a five-level scale with 0.69 accuracy (Section 5.2), and that errors are predominantly local, typically involving an adjacent level rather than a confusion between a disinformation site and an established newspaper. We claim no improvement in information quality itself: the intended effect is confined to allocating expert effort.

Preconditions and Their Plausibility Four conditions must hold, and they are unevenly likely to. **The ground truth must be valid**: we inherit the NewsGuard scale together with its editorial judgements, and nothing internal to our data can test this. **The unit must be the publisher**, never the article: article-level accuracy is 0.60 and supports no claim about an individual piece. **Enough text must be available**, so that small or recent outlets — those most likely to

lack a rating — are the least likely to qualify. **Use must remain advisory**, ordering a review queue rather than rating anything autonomously. Our evidence, finally, covers English-language political news alone.

Negative Externalities and Responsible Deployment The principal risk is measured rather than hypothesised. Misclassification rates vary systematically with political leaning (Section 5.4), reaching 59–72% for far-left and far-right outlets against 30–34% for slightly right and non-labelled ones, and reliability is overestimated at either extreme while underestimated for slightly left-leaning outlets. Such error is systematic and reproducible, which distinguishes it in kind from that of an individual rater. Two further risks follow from presentation and use: a single level out of five reads as a measurement although the l_4/l_5 boundary is close to unresolvable (Section 5.3), and an inexpensive score is a target, optimisable on surface features. We therefore regard the following as necessary rather than advisable: publishing per-leaning error rates; abstaining when few articles are available; reporting an interval rather than a single level; and confining the output to ordering human review. The bias audit of Section 5.4 is constitutive of the contribution, not supplementary to it.

7 Discussion and Conclusion

In this study, we investigated whether the reliability of news publishers could be inferred from their articles, framing the task as a five-level ordinal classification problem aligned with NewsGuard ratings and evaluating four transformer-based models under a strict publisher-disjoint protocol.

Our findings demonstrate that language models can capture meaningful and generalizable signals of publisher reliability. Interestingly, the substantially larger Mistral 24B does not outperform smaller architectures, suggesting that performance is limited by dataset characteristics rather than model size. Despite operating with a 1,024-token context window, four times larger than the limit used for the other models, Mistral shows no measurable gain, which may indicate that the cues discriminating reliable publishers are concentrated in the beginning of articles and further supports the hypothesis that increasing model size or input length alone cannot overcome limitations imposed by weak supervision. A key finding is that publisher reliability should not be modelled as a property of isolated articles: it emerges as a collective signal that becomes more stable when evidence is aggregated across a source, which supports a shift from article-centric formulations toward source-level inference. The scope constraints and the risks of the approach are set out in Section 6.

Future research should extend this framework to multilingual settings, explore confidence-aware aggregation strategies, and develop training objectives tailored to ordinal classification. Combining content-based approaches with ecosystem-level signals may also provide a more robust framework for scalable source assessment.

Disclosure of Interests. The authors have no competing interests to declare.

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